Tungsten Carbide

01 What is hiding at the end of a pen?



Many ballpoint pens have a tiny tungsten-carbide ball.

Look closer. Look for the tiny round ball where ink meets paper.

A little explanation. The tiny ball in many ballpoint pens is tungsten carbide. It rolls ink onto paper and resists wear. People make this material by joining tungsten with carbon, then pressing powder and heating it very hot. This heating step is called sintering. Usually, a little cobalt metal binds the grains together: cemented carbide.

Try this. Trace the rolling ball in the picture, then point to the line it leaves.

02 What can a scratch tell us?



A harder object can leave a scratch in a softer one.

Look closer. Big moves the craft stick across soap on a tray. Little watches and records.

A little explanation. A harder thing can scratch a softer thing. Hard means resisting scratches and dents. Tungsten carbide is one of the hardest materials people make. It is also heavy for its size. The soap test lets Little see a surface give way.

Try this. Predict what mark Big's craft stick will leave in the soap. Record what you notice.

03 Does hard always mean tough?



Chalk can snap. A rubber eraser can bend back.

Look closer. Compare the chalk pieces with the eraser bending in Big's hands.

A little explanation. Tough means resisting cracks or breaks when bumped, bent or dropped. Hard and tough are different. Chalk is brittle: it snaps when bent. An eraser is tough: it bends back. Tungsten carbide is very hard but less tough than steel. It is brittle enough to chip if knocked hard.

Try this. Say which test gives a clue about a surface, and which gives a clue about breaking.

04 Why give a hard tip a tough body?



Choose a material for the job it needs to do.

Look closer. Little's sketch joins a small tip to a larger body.

A little explanation. Engineers often join a carbide tip to a tougher steel body, as on saw teeth and some masonry drill bits. The tip resists wear while the steel takes bumps. Carbide also appears in some ski-pole and hiking-pole tips and ice-grip studs. Choose a material for its job: hard, tough, or both.

Try this. Choose a job for the tip, then a job for the body.